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Mutual Space Representation Standardization for Mixed and Augmented Reality Remote Collaboration

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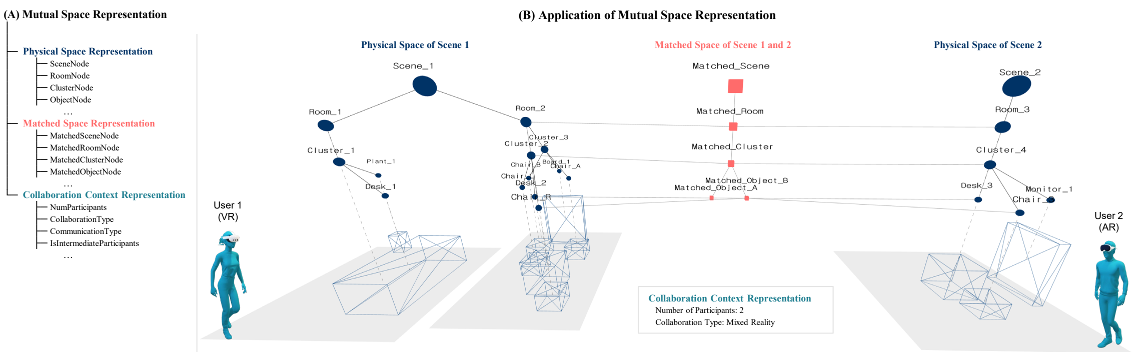

Figure 1: Figure 1 illustrates (A) the hierarchical structure of the proposed mutual space representation and (B) an application scenario. In this example, two users collaborate in remote environments: User 1 in Virtual Reality (VR) and User 2 in Augmented Reality (AR). Scene 1, associated with User 1, consists of two rooms and three clusters, whereas Scene 2, associated with User 2, consists of one room and one cluster.

Abstract

This paper proposes a mutual space representation to support remote collaboration in Mixed and Augmented Reality (MAR) environments. To achieve high immersion and coexistence in MAR remote collaboration, a mutual space based on heterogeneous physical spaces that can be commonly utilized by collaborating participants is required. Although previous studies have proposed various spatial matching and modifying methods to overcome spatial heterogeneity, their different representations have made their integrated use in real-world scenarios difficult. To overcome these limitations, this paper proposes a mutual space representation standard that includes the representation of physical space, matching space, and collaboration context required for mutual space generation and management. We expect that the proposed standards will serve as a foundation for seamlessly applying solutions accumulated in academia to industrial systems, thereby providing users with a more immersive and coexistent MAR remote collaboration experience.

Index Terms: Mixed Reality, Augmented Reality, Virtual Reality, Remote Collaboration, Mutual Space, Standardization

1 Introduction

As the need for effective communication and collaboration among remote users increases, various technologies have been developed to support remote collaboration. Mixed and Augmented Reality (MAR) allows users to meet and interact together in a virtual space through avatars. In particular, in MAR remote collaboration, mutual space, a virtual shared space that reflects the physical space, is required to make users feel as if they are in the same local space. Mutual space is generated by reflecting common spatial elements between different physical spaces. It is a key element that provides a high sense of coexistence and realism by allowing them to share the same interactions and behaviors through mutual space in MAR remote collaboration.

The biggest challenge of mutual space is to solve heterogeneity between spaces and maximize common spatial elements. To this end, various methods have been studied to match common objects or spatial structures in each physical space [5]. In particular, there have been recent studies that match and align similar elements using semantic information in addition to geometric information [9, 6, 4]. In addition, in spaces with high heterogeneity, attempts have been made to move avatars or adjust and transform virtual spaces [19, 8]. However, since each study focused on different spatial information, it was difficult to integrate and flexibly utilize the methods in a real collaboration scenario.

This paper proposes a mutual space representation standard to overcome the limitation that it is difficult to integrate due to the different types and representations of spatial elements used. The representation standard consists of three parts, considering the holistic process of mutual space generation. The first physical space representation presents a standard for representing input physical spaces with geometric and semantic information. The second matched space representation presents a standard for representing spatial elements that match given physical spaces. The last collaboration context representation contains contextual information, including collaboration situations or scenarios.

The proposed standard will play a role in bridging the gap between academia and industry in the generation and management of mutual spaces in MAR remote collaboration. It will contribute to increasing the possibility that the methods in academia can be applied as solutions in the industry. In addition, the industry will reduce the cost of implementing methods as systems or services, through this standard. Users will feel a high sense of immersion and realism in MAR remote collaboration through services implemented with a unified mutual space representation. This virtuous

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cycle structure is expected to contribute to the increase in MAR remote collaboration users and the growth of related industries by contributing to the spread of academic methods to users.

2 Related Works

2.1 Mutual Space Generation Method

The biggest goal of mutual space creation and management is to configure an environment where remote participants can meet and interact. To form a mutual space, it is essential to find common spatial elements or areas of two physical spaces and match them [10, 13]. However, high heterogeneity between physical spaces in the real world was the biggest challenge in remote collaboration research, which decreases applicability on diverse real spaces [3]. To this end, many studies have matched various spatial elements by integrating and utilizing not only geometric information of spaces and internal objects but also semantic information [2, 9, 6]. Also, methods to match desks or boards as certain single objects and maximize their utilization have been studied [2, 1]. In addition, methods for adjusting and transforming the movement of perceived spaces or avatars have been proposed to address the very high heterogeneity between spaces caused by the diversity and complexity of real spaces [5, 7, 8, 19, 18]. However, the proposed methods have limitations in that they are difficult to utilize in an integrated and fluid manner because they each propose and use their own spatial data representations. Therefore, a unified spatial representation is required for various methods to be used organically and complexly in real-world spaces and collaboration scenarios.

2.2 Space and Scene Representation

There have also been studies that have attempted to manage space or scenes by representing data. In particular, studies have been conducted to create and synthesize new virtual spaces by representing scenes as graph structures [15, 16, 17, 20, 12]. There have also been studies that have attempted to track long-term changes in space by utilizing the variability of space [11]. However, the purpose of these spatial representations is to create plausible virtual scenes that do not exist in reality or to manage scenes as digital twins of reality. These are limited not only in handling contextual data but also in considering the experience of space. Because the spatial characteristics of the environment affect virtual or augmented reality experiences [14], it is necessary to represent spatial data of interpreted properties and structured relationships for remote collaboration from the perspective of spatial experience. Therefore, a new standardized space representation is needed to generate and manage mutual spaces for an immersive and realistic MAR remote collaboration experience.

3 Mutual Space Representation

Mutual space representation consists of three major parts: Physical space representation, matched space representation, and collaboration context representation. First, physical space representation deals with how to represent the physical spaces where remote collaboration participants are located. Second, matched space representation contains how to represent the matched space elements between these spaces. Finally, collaboration context representation is for context information under a collaboration scenario and situation, which should be considered when generating and managing mutual space. These three parts suggest that when forming mutual space, not only the geometric and semantic information of the physical space but also the collaboration information should be considered in an integrated manner. The details are as follows.

3.1 Physical Space Representation

Physical space representation is a hierarchical graph structure designed to systematically transfer real environments into digital data, considering mutual space generation and management. This structure consists of four main node types: scene, room, cluster, and object from top to bottom. A scene node represents the entire space, and it contains room nodes to express physical sections divided by walls. A room can contain cluster nodes grouped by the interrelation between objects, and each cluster contains object nodes that can actually interact. All nodes commonly include an ID for identification, a label, parent-child relationship information, and geometric properties such as position, rotation, and size. In particular, object nodes have additional properties such as affordance (e.g., sittable, grabbable, openable), virtuality (distinguishing between physical and virtual objects), variability (possibility of change), and interaction direction (direction of objects that the user can interact with). These unique properties are used to determine the interrelation between objects.

Edges connect these nodes and specifically define the relationships between spatial elements. They are broadly divided into vertical edges and horizontal edges. A vertical edge connects the upper and lower level nodes to clearly show the hierarchical structure. For example, the relationship between a scene and a room, or a room and a cluster, is modeled as a vertical edge to express the containment relationship within the space. A horizontal edge represents the relationship between nodes at the same hierarchical level, and can express physical proximity and functional connectivity. For example, it can indicate that two clusters in the same room are close to each other or are functionally connected, and it can also express the connectivity between objects in the same cluster. A horizontal edge can include a weight attribute that indicates the strength or importance of the relationship between two elements. This node and edge-based hierarchical graph representation elaborates how spatial elements are connected and interact with each other, and serves as the basic framework for subsequent matching and modifying physical space data.

3.2 Matched Space Representation

Matched space representation is a structure designed to connect two or more different spatial elements in each space that are semantically related. Basically, if elements have the same physical layout and geometry, they can be matched easily. Furthermore, elements that have the same function or purpose can be grouped into a single matched node, even if their physical layout or geometry is not the same. For example, desks in different physical spaces may have different sizes or local positions, but they can be represented as a matched object node because they share the same functional meaning of a work surface; in other words, they share the same affordance. This matching not only considers geometrical matching but also considers the semantic matching of each element, allowing remote participants to interact in a single shared context, even when the physical spaces are highly heterogeneous.

Matched nodes also have a hierarchical structure similar to the nodes in the physical space representation. Additionally, each matched node contains a list of matched information that clearly describes which scene and which source node it refers to. For example, a matched object node can connect a conference chair in one space to a stool in another space, indicating that different types of objects perform the same 'sittable' affordance. A matched node is identified by its own ID and label, and can contain properties that are inherited or summarized from the source node (e.g., affordance, virtuality). Concrete application examples of matched space representation are given in section 4.

3.3 Collaboration Context Representation

Collaboration context representation goes beyond using spatial data as a set of objects and includes context that considers how users achieve spatial experience in the mutual space during MAR remote collaboration. Even if spaces are represented and matched, the interpretation and use of the space can vary depending on the actual collaboration scenario. The layout and emphasis of the space can vary depending on whether there are many or few participants, whether the interaction in collaboration is symmetrical or asymmetrical, and whether the primary communication method is voice, gesture, or text. If this contextual information is provided together, it can provide a spatial representation optimized for an actual collaboration scenario.

This representation includes several properties that help understand the context of collaboration. NumParticipants indicates the number of people currently participating in the collaboration which directly affects how the space is divided. CollaborationType describes the device and environment of users to distinguish whether the interaction possibilities are symmetrical or asymmetrical, allowing different space generation and management strategies to be applied. CommunicationType specifies the communication method (voice, gesture, text, etc.) used by each participant, allowing the system to consider a spatial interface or interaction method that suits that method. Finally, IsIntermediateParticipants records whether there are intermediate participants, so it determines optimization of mutual space through regeneration or remodifying the current mutual space. By combining these contextual properties, spatial representations provide meaningful frameworks that reflect users' spatial experience in remote collaboration.

4 Application Scenario

We will explain the process of applying the proposed space representation standard to an example collaboration scenario as a case study(Figure 1). We assume a MAR remote collaboration scenario where two different physical spaces are connected to allow remote collaboration. The first scene (Scene 1) and the second scene (Scene 2) represent different physical scenes, and each scene contains rooms, clusters, and various objects arranged hierarchically. Scene 1 has two rooms; the first room has a cluster with a work desk and decorative plants. Another room has clusters, one with a desk and chairs on both sides, so that users can place objects or sit and collaborate. In another cluster, two chairs face the board, so that participants can interact by writing on the board or sharing content while sitting. In the room of Scene 2, a desk, chair, and monitor are gathered in one cluster, so that users can place objects or sit and work, and observe or present visual materials through the monitor. As the collaboration context data, there are two participants, and the collaboration is conducted without an intermediate participant. The collaboration type is a heterogeneous VR–AR collaboration, since User 1 is using an Augmented Reality (AR) device while User 2 is using a Virtual Reality (VR) device. In this way, each space has objects that support various interactions, such as placing objects, walking, sitting, writing, and observing. These are interrelated and grouped at a higher level of nodes in the representation hierarchy.

Nodes in each space that are semantically and functionally corresponding are matched to allow a common interaction. This matching is not fixed; it can be achieved in various ways. For example, if structural information at the room level is matched, locations can be shared based on the same room section, even if they are in different spaces. Participants can be identified in real time to determine which room they are active in and coordinate spatial movement by aligning the coordinates. In this case, methods that can maximize the utilization range of the floor or walls of the room can be applied [7]. In another way, matching can be extended to the level of clusters and objects within them. If a single object, such as a desk, is matched, documents can be placed and reviewed on the same desk, and different research methods can be used to maximize the use of the matched object, depending on its type [2, 8]. In addition, if two or more objects are used simultaneously, complex interactions can be shared depending on which objects are matched. For example, if a board and chairs facing it are matched, users can collaborate by sitting on a chair and writing down ideas on the board. In another combination, a desk and a chair can provide a realistic experience as if users were sitting across the same table and discussing or exchanging tools, even though users are in different spaces. In this scenario, methods that support a dynamic collaboration scenario can be utilized [9, 1]. In this way, the information of the space that is matched varies depending on the characteristics and composition of physical space, and the matching method that can be utilized also varies. As a result, the scope and depth of interaction allowed in the mutual space vary, and through this, participants can share a collaborative experience similar to being together in the same physical space.

There are also various possibilities when implementing the mutual space representation standard as a system or service application in the industry. Depending on the collaboration situation that each application aims for, the type of collaboration task that must be performed, the maximum number of participants, the type of device users use, and whether intermediate participants are allowed, the priority of which matching information in the physical space and which method to apply will vary. For example, in a collaborative system involving many people, the focus may be on structural matching at the room level to maximize the possible movement range. Whether users are wearing an AR or VR device can affect their interaction possibilities because of the different specs of the device, and the mutual space may support them. MAR remote collaboration application in the industry should consider the complex collaboration situation. Therefore, a common spatial representation that enables these various methods to be applied in an integrated or adaptive manner is required. This standard ensures data compatibility between systems and provides a foundation for applying various expansion scenarios. Ultimately, by utilizing this standard, the industry will be able to implement more practical collaborative experiences by flexibly generating mutual spaces according to different collaborative needs and environments.

5 Conclusion

The mutual space representation proposed as a standard in this paper is a space representation for the integrated use of a series of processes that match common space elements between each physical space of remote participants and transform collaboration elements to maximize them. The physical space representation, matched space representation, and collaboration context representation that reflect the process of mutual space generation and management reflect the various research methods developed in academia to overcome heterogeneity between spaces. This provides a foundation for the proposed methods to be utilized more extensively and comprehensively in real environments, helping participants experience remote collaboration based on physical spaces in various real-world environments. In the industry, it opens the way for technologies in the lab to be expanded into actual products and services by reducing the cost in the system development process caused by different spatial representations of academic solutions. This standard will be continuously verified and improved through various real-world scenarios, and will serve as a bridge between academia and industry, ultimately contributing to delivering high immersion and coexistence to users in MAR remote collaboration.

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REFERENCES

[1] J. E. S. Grønbæk, K. Pfeuffer, E. Velloso, M. Astrup, M. I. S. Pedersen, M. Kjær, G. Leiva, and H. Gellersen. Partially blended realities: Aligning dissimilar spaces for distributed mixed reality meetings. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2023. 2, 3

[2] J. E. S. Grønbæk, J. Sánchez Esquivel, G. Leiva, E. Velloso, H. Gellersen, and K. Pfeuffer. Blended whiteboard: Physicality and reconfigurability in remote mixed reality collaboration. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2024. 2, 3

[3] D. Jo, K.-H. Kim, and G. J. Kim. Spacetime: adaptive control of the teleported avatar for improved ar tele-conference experience. *Computer Animation and Virtual Worlds*, 26(3-4):259–269, 2015. 2

[4] M. Keshavarzi, A. Y. Yang, W. Ko, and L. Caldas. Optimization and manipulation of contextual mutual spaces for multi-user virtual and augmented reality interaction. In *2020 IEEE conference on virtual reality and 3D user interfaces (VR)*, pp. 353–362. IEEE, 2020. 1

[5] D. Kim, H.-I. Kim, and W. Woo. Mutual space generation with relative translation gains in redirected walking for asymmetric remote collaboration. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 674–677. IEEE, 2022. 1, 2

[6] D. Kim, S. Kim, S. Choi, and W. Woo. Spatial affordance-aware interactable subspace allocation for mixed reality telepresence. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 1256–1265. IEEE, 2024. 1, 2

[7] D. Kim, J.-e. Shin, J. Lee, and W. Woo. Adjusting relative translation gains according to space size in redirected walking for mixed reality mutual space generation. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 653–660. IEEE, 2021. 2, 3

[8] D. Kim and W. Woo. Edge-centric space rescaling with redirected walking for dissimilar physical-virtual space registration. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 829–838. IEEE, 2023. 1, 2, 3

[9] S. Kim, D. Kim, J.-E. Shin, and W. Woo. Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 796–805. IEEE, 2024. 1, 2, 3

[10] N. H. Lehment, D. Merget, and G. Rigoll. Creating automatically aligned consensus realities for ar videoconferencing. In *2014 IEEE international symposium on mixed and augmented reality (ISMAR)*, pp. 201–206. IEEE, 2014. 2

[11] S. Looper, J. Rodriguez-Puigvert, R. Siegwart, C. Cadena, and L. Schmid. 3d vsg: Long-term semantic scene change prediction through 3d variable scene graphs. *arXiv preprint arXiv:2209.07896*, 2022. 2

[12] R. Ma, A. G. Patil, M. Fisher, M. Li, S. Pirk, B.-S. Hua, S.-K. Yeung, X. Tong, L. Guibas, and H. Zhang. Language-driven synthesis of 3d scenes from scene databases. *ACM Transactions on Graphics (TOG)*, 37(6):1–16, 2018. 2

[13] S. Orts-Escolano, C. Rhemann, S. Fanello, W. Chang, A. Kowdle, Y. Degtyarev, D. Kim, P. L. Davidson, S. Khamis, M. Dou, et al. Holoportation: Virtual 3d teleportation in real-time. In *Proceedings of the 29th annual symposium on user interface software and technology*, pp. 741–754, 2016. 2

[14] J.-E. Shin, B. Yoon, D. Kim, H.-I. Kim, and W. Woo. The effects of device and spatial layout on social presence during a dynamic remote collaboration task in mixed reality. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 394–403. IEEE, 2022. 2

[15] H. Wang, W. Liang, and L.-F. Yu. Scene mover: automatic move planning for scene arrangement by deep reinforcement learning. *ACM Transactions on Graphics (TOG)*, 39(6):1–15, 2020. 2

[16] K. Wang, Y.-A. Lin, B. Weissmann, M. Savva, A. X. Chang, and D. Ritchie. Planit: Planning and instantiating indoor scenes with relation graph and spatial prior networks. *ACM Transactions on Graphics (TOG)*, 38(4):1–15, 2019. 2

[17] Q. A. Wei, S. Ding, J. J. Park, R. Sajnani, A. Poulenard, S. Sridhar, and L. Guibas. Lego-net: Learning regular rearrangements of objects in rooms. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19037–19047, 2023. 2

[18] L. Yoon, D. Yang, C. Chung, and S.-H. Lee. A full body avatar-based telepresence system for dissimilar spaces. *arXiv preprint arXiv:2103.04380*, 2021. 2

[19] L. Yoon, D. Yang, J. Kim, C. Chung, and S.-H. Lee. Placement retargeting of virtual avatars to dissimilar indoor environments. *IEEE Transactions on Visualization and Computer Graphics*, 28(3):1619–1633, 2020. 1, 2

[20] S.-K. Zhang, W.-Y. Xie, and S.-H. Zhang. Geometry-based layout generation with hyper-relations among objects. *Graphical Models*, 116:101104, 2021. 2